Ex.No.4 – Generate the Prompt for Different Prompt Types

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Aim

To design and generate prompts for the following prompt types used in prompt engineering:

1. Comparative Analysis Prompt
2. Experiential Perspective Prompt
3. Everyday Functioning Prompts

The objective of this experiment is to understand how different prompt types guide artificial intelligence systems to produce structured, meaningful, and context-specific responses.

Introduction

Prompt engineering is the process of designing effective inputs or instructions that guide an Artificial Intelligence (AI) system to produce accurate and useful responses. The quality of the prompt directly influences the quality of the output generated by AI models. By using different types of prompts, users can control how AI analyzes information, explains concepts, or solves problems.

Different prompt styles are used depending on the type of response required. For example, comparative prompts help in analyzing differences between multiple options, experiential prompts help in understanding a concept from a particular viewpoint, and everyday functioning prompts explain how technologies work in daily life situations.

In this experiment, we generate prompts based on three prompt types and apply them to a practical scenario related to choosing a smartphone for daily use.

Procedure

Step 1: Define the Scenario and Use Case

Scenario:
You are planning to purchase a new smartphone for your daily activities. The smartphone should support a **long battery life**, a **good AI-powered camera for photography**, and **smooth performance** for common activities such as social media usage, online classes, video streaming, messaging, and entertainment.

However, there are many smartphone models available in the market, and you are confused about which one to choose. You want to compare two smartphone models to determine which device offers better features and value for money.

At the same time, you also want to understand how features such as battery optimization, AI cameras, and high performance processors help users in their everyday life.

Therefore, prompts are designed to analyze, understand, and explain these aspects using different prompt engineering approaches.

Step 2: Identify Prompt Patterns for Each Prompt Type

1. Comparative Analysis Prompt

A **Comparative Analysis Prompt** is used when we want the AI system to compare two or more concepts, products, or ideas. The AI analyzes their similarities, differences, advantages, and disadvantages based on certain criteria.

This type of prompt is useful for decision-making situations such as choosing between products, comparing technologies, or evaluating different methods.

Comparative prompts typically include:

- Multiple options to compare
- Specific criteria for comparison
- A request for conclusion or recommendation

Prompt Example (Smartphone Use Case)

"Compare **iPhone 15** and **Samsung Galaxy S23** in terms of battery life, AI camera features, performance, storage capacity, display quality, and price. Identify the key similarities and differences between the two smartphones and explain which device would be more suitable for daily activities such as communication, social media browsing, photography, and entertainment."

Additional Example

"Compare **electric vehicles and petrol vehicles** in terms of environmental impact, maintenance cost, fuel efficiency, and long-term sustainability. Explain the advantages and disadvantages of each type of vehicle."

Another Example

"Compare **machine learning and deep learning** in terms of working principles, complexity, training data requirements, and real-world applications."

These prompts encourage the AI to produce structured comparisons that help users make informed decisions.

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2. Experiential Perspective Prompt

An **Experiential Perspective Prompt** asks the AI to explain a concept from the viewpoint of a particular person, profession, or role. This type of prompt helps generate responses that reflect real-life experiences, opinions, and perspectives.

By framing the prompt from a specific role such as a student, teacher, engineer, or business owner, the AI can provide more relatable and practical explanations.

This prompt type is commonly used in:

- Education
- Workplace scenarios
- Professional insights
- Personal experiences

Prompt Example (Smartphone Use Case)

"From the perspective of a college student, explain how a smartphone with long battery life, an AI-powered camera, and smooth performance can support daily activities such as attending online classes, completing assignments, communicating with classmates, capturing photos, and managing study schedules."

Additional Example

"As a software engineer, describe how cloud computing platforms help improve software development, data storage, scalability, and collaboration among team members."

Another Example

"From a teacher's perspective, explain the benefits and challenges of using online learning platforms for teaching students."

This type of prompt helps the AI generate responses that are **experience-based and context-aware**, making them easier for users to understand.

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3. Everyday Functioning Prompts

An **Everyday Functioning Prompt** asks the AI to explain how a particular technology, concept, or system works in everyday life. These prompts focus on practical applications and real-world usage.

They are useful for explaining complex technologies in a simple and relatable manner.

This prompt type is commonly used to explain:

- Artificial intelligence applications
- Smartphone features
- Internet technologies
- Security systems
- Digital services

Prompt Example (Smartphone Use Case)

"Explain how AI-powered cameras and battery optimization features in smartphones improve everyday activities such as taking high-quality photos, making video calls, browsing social media, and watching videos for long periods without frequent charging."

Additional Example

“Describe how GPS technology in smartphones helps people in daily activities such as navigation, tracking food delivery, locating nearby services, and using ride-sharing applications.”

Another Example

“Explain how recommendation algorithms are used in applications such as YouTube and Netflix to suggest movies and videos based on user preferences.”

These prompts help users understand **how modern technologies influence daily life and improve user experiences**.